University of Connecticut | Master of Science in Financial Risk Management

FNCE 5352 – Financial Programming and Modeling
January 20, 2026

1. Course Description:

This course will introduce the students to a wide variety of algorithms that are used in machine learning applications. The course emphasizes the core ideas of statistical learning, including regression, classification, model assessment, and the trade-offs involved in building predictive models. Students will code a few algorithms completely and learn to use software packages that implement others, with an emphasis on understanding model behavior, interpretation, and performance evaluation. Instruction and assignments will use the R and Python programming ecosystems. Students will be exposed to Machine Learning at scale using the Sklearn, Keras, TensorFlow, and PyTorch libraries. Throughout the course, special attention will be given to applications of these algorithms to finance.

2. Instructors

Ed Hayes: ehayes@uconn.edu

Matt McDonald: matthew.mcdonald@uconn.edu

3. Course Delivery

14 lectures

4. Academic Integrity

Students must adhere to the University of Connecticut Student Code, which can be found at: https://community.uconn.edu/the-student-code-pdf/. Additionally students must abide by the Academic, Scholarly, and Professional Integrity and Misconduct Policy Assignments and/or quizzes must be completed individually.

5. AI Usage

The use of AI tools (such as large language models and coding assistants) is **permitted and encouraged** in this course. Developing the ability to work effectively with AI is an increasingly important professional skill, particularly in data-driven and finance-related roles, and students are expected to engage with these tools thoughtfully. Use of AI is **not factored into grading**, either positively or negatively. The only requirement is transparency: each assignment submission must include a brief statement indicating **whether AI tools were used, and if so, how** (for example, idea generation, code debugging, explanation, or drafting). Students remain fully responsible for the accuracy, originality, and integrity of all submitted work, regardless of whether AI tools were used.

6. Required Texts

Part 1 (R and RStudio, with Professor McDonald)

We will be using the book "R for Data Science" by Hadley Wickham and Garrett Grolemund as the primary reference for working with data in R, including data wrangling, visualization, and exploratory data analysis. The book is available online at https://r4ds.hadley.nz/. It is free and licensed under the CC BY-NC-ND 3.0 License If you'd like a physical copy of the book, you can order it on Amazon.

In addition, we will use the book *An Introduction to Statistical Learning* by Gareth James, Daniela Witten, Trevor Hastie, and Robert Tibshirani as a primary conceptual reference for statistical learning methods, including regression, classification, model assessment, and regularization. A free online version of the book is available[1] at https://www.statlearning.com/. We will refer to this book as ISLR and ISLP.

The book "Feature Engineering and Selection" by Max Kuhn and Kjell Johnson will be used selectively as a supplemental reference, particularly for discussions related to feature construction and practical modeling considerations. Physical copies are sold by **Amazon** and **Taylor & Francis**. An online version is available at https://bookdown.org/max/FES/. Students are not expected to read this book cover-to-cover.

[1] The website has versions with R code and Python code. We will refer to the R version during the first half of the course and the Python version during the second half. We refer to them as ISL-R and ISL-P.

Part 2 (Python, with Professor Hayes)

As noted above, we will also use *An Introduction to Statistical* Learning in Part 2 of the course.

Listed below are four books that we have used in past versions of this course. I do NOT recommend you buy these books. The applications of ML and AI to finance have been rapidly evolving and we want to reflect that in this course. Book 3 (Stefan Jansen) was the primary book for several years, but we moved from it when the Python notebooks became outdated. A new edition is coming out, and we hope to be able to use it in Part 2.

1) Tatsat, Hariom, *et al*. *Machine Learning and Data Science Blueprints for Finance: from Building Trading Strategies to Robo-Advisors Using Python*. O'Reilly Media, 2020.
2) Tony Guida, *et al*. *Big Data and Machine Learning in Quantitative Investment*. Wiley, 2019.
3) Stefan Jansen, *Machine Learning for Algorithmic Trading: Predictive models to extract signals from market and alternative data for systematic trading strategies with Python, 2nd Edition*. Packt Publishing, 2021
4) Géron, Aurélien. *Hands-on Machine Learning with Scikit-Learn, Keras, and TensorFlow: Concepts, Tools, and Techniques to Build Intelligent Systems*, 3e. O'Reilly, 2022.

6. Part 1 (R and RStudio) Prerequisites

Course Materials

Course materials can be found at https://github.com/mattmcd71/fnce5352_spring2026. This repository will be updated throughout the semester and will contain lecture materials, code examples, and assignment instructions.

R Fundamentals

The course assumes some understanding of the R programming language. If you would like to get a basic introduction to the R programming language, please visit the following link:

https://support.posit.co/hc/en-us/articles/201141096-Getting-Started-with-R

R Installation

We will be periodically using R and RStudio interactively during the class instruction. If you would like to follow along during the class, please follow the instructions below.

Local Installation Instructions:

R

We'll be using the most recent version of R locally but but any reasonably recent version (greater than 3.5.1) should be fine.

R can be downloaded from the following link: https://www.r-project.org/

RStudio

RStudio is an Interactive Development Environment for the R programming language. It is very useful. You can download it at:

https://posit.co/download/rstudio-desktop/

Positron is a newer integrated development environment from Posit that supports both R and Python workflows. While course instruction will continue to use RStudio, students who are interested are welcome to explore Positron on their own. Use of Positron is optional and will not be required for the course.

7. Part 2 (Python) Prerequisites

If you want to get a head start on the Python section, you can do the following:

1. We will be covering the topics in Chapters 8 and 12 of ISLP, so you can read these sections if you want to do some thinking about the topics.

2. If you want to work on Python programming in the concept of Machine Learning, there are sections on Python programming at the end of each chapter in ISLP. These sections will help you to improve your Python programming and deepen your knowledge of ML topics at the same time.

8. Lessons and Assignments

Assignments during the semester will emphasize applied analysis and interpretation of results using real financial data. Specific assignment details may be adjusted as the semester progresses.

In Part 2, the reading assignments are left as TBD pending the publication of updated editions of some of the books listed above for Part 2.

Lecture Date	Topic	Assignment	Reading assignment (before next class)
PART 1			
20-Jan	Intro to R, RStudio and the Data Science Workflow		R4DS: Sections 1-8 ISL: Chapter 1 (conceptual overview) Articles: “Good enough practices in scientific computing” “Excuse me, do you have a moment to talk about version control?”

27-Jan	Analytic Workflow & Visualization	R4DS: 1.2.5: 1, 3, 4 1.4.3: 1 1.5.5: 1, 2, 3	R4DS: Sections 9-11 ISL: Chapter 2 (why we estimate models; train vs test error)
3-Feb	Data Wrangling, Tidy Data, and Exploratory Analysis	R4DS: 3.2.5: 1, 2, 4 3.3.5: 1	R4DS: Sections 12-19 ISL: Chapter 3 (sections on simple linear regression – conceptual)
10-Feb	Linear Regression: Introduction and Interpretation		R4DS: Sections 25-27 ISL: Chapter 3 (simple and multiple linear regression) SFR Rent vs Buy Analysis
17-Feb	Model Assessment and Resampling Methods		ISL: Chapter 5 (validation sets, cross-validation, bootstrap)
24-Feb	Multiple Regression, Qualitative Predictors, and Diagnostics	SFR Rent v Buy Analysis Due	ISL: Chapter 3 (qualitative predictors and diagnostics)
3-Mar	Classification and Logistic Regression	Credit Modeling Project (due end of semester)	ISL: Chapter 4 (logistic regression and classification)
PART 2			
	Topics	**Case Studies**	**Readings**
10-Mar	Machine Learning in Finance: Modeling Perspective and Framework; Regression and Classification	Stock Price Prediction; Yield Curve Prediction	TBD
24-Mar	Decision Trees, Random Forests, and Gradient Boosting	Fraud Detection; Default Prediction	TBD

31-Mar	Neural Networks, Time Series Modeling, LSTM	Stock Price Prediction; Bitcoin Trading	TBD
7-Apr	Unsupervised Learning: PCA and Kernel PCA	Eigen Portfolios; Yield Curve Modeling; Bitcoin Trading	TBD
14-Apr	Unsupervised Learning: t-SNE; Clustering Techniques	Pairs Trading; Hierarchical Risk Parity	TBD
21-Apr	Reinforcement Learning	RL-based Trading Strategy	TBD
28-Apr	Reinforcement Learning	Portfolio Allocation	TBD